

SOFTWARE REQUIREMENTS SPECIFICATION

Health Management System

Document Title	Software Requirements Specification — HMS
Version	1.0
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Date	15 March 2026
Status	Final Draft

Table of Contents

1. Introduction

- 1.1 Purpose
- 1.2 Document Conventions
- 1.3 Intended Audience
- 1.4 Project Scope
- 1.5 Definitions & Acronyms

2. Overall Description

- 2.1 Product Perspective
- 2.2 Product Functions
- 2.3 User Classes
- 2.4 Operating Environment
- 2.5 Constraints & Assumptions

3. Functional Requirements

- 3.1 Patient Management
- 3.2 Doctor Management
- 3.3 Appointment Scheduling
- 3.4 Medical Records
- 3.5 Billing & Inventory
- 3.6 Admin Dashboard
- 3.7 Reporting Module

4. External Interface Requirements

5. Non-Functional Requirements

- 5.1 Performance
- 5.2 Security
- 5.3 Usability
- 5.4 Reliability
- 5.5 Scalability

6. Conclusion

1. Introduction

1.1 Purpose

This SRS describes complete requirements for the Health Management System (HMS) v1.0 — a web-based application designed to help hospitals and clinics manage patient information, doctor details, appointments, medical records, and billing efficiently. This system reduces manual work and improves the management of healthcare services.

1.2 Document Conventions

- SHALL — mandatory requirement
- SHOULD — recommended requirement
- MAY — optional requirement

1.3 Intended Audience

Audience	Relevant Sections
Project Developers / Programmers	All sections — especially Sections 3 and 5
Project Supervisor / Evaluator	Sections 1, 2, and 5
UI/UX Designers	Section 4 and Section 3 (feature flow)
Testing / QA Team	Sections 3 and 5
Hospital / Clinic End Users	Sections 2 and 3

1.4 Project Scope

The HMS provides role-based portals for Admins, Doctors, and Patients. Excluded from v1.0: Telemedicine video, third-party insurance gateway, native mobile apps.

- Patient registration, profile management, and health history tracking
- Doctor profile management and availability scheduling
- Appointment booking, modification, and cancellation
- Electronic Health Record (EHR) creation and retrieval by doctors
- Automated billing, invoice generation, and payment tracking
- Inventory monitoring and low-stock alerting
- Analytics and custom report generation (PDF/CSV export)
- Secure, role-based Admin control panel

1.5 Definitions and Acronyms

Term	Description
HMS	Health Management System
Admin	Administrator — full system access
Patient	Person receiving medical treatment

Term	Description
Doctor	Medical professional providing treatment
EHR	Electronic Health Record — digital patient chart
JWT	JSON Web Token — authentication token
RBAC	Role-Based Access Control

2. Overall Description

2.1 Product Perspective

The HMS is a standalone web-based application using a three-tier architecture:

Tier	Technology
Frontend (Presentation)	HTML5, CSS3, JavaScript (ES6+)
Backend (Application)	Node.js v18+ with Express.js REST API
Database (Data)	MongoDB 6.x (primary) / MySQL 8.x (optional)

2.2 Product Functions

- Patient registration and login
- Doctor registration and management by Admin
- Appointment booking with real-time availability check
- Viewing and updating patient medical history (EHR)
- Billing and payment management with automated invoices
- Admin dashboard for complete system management
- Reporting and data export in PDF/CSV formats

2.3 User Classes and Characteristics

User Class	Description	Access Level
Administrator	Hospital IT staff managing config, accounts, settings	Full access — all modules
Doctor	Medical professionals — view/update EHRs, manage schedule	EHR, appointments, schedule
Patient	Registered individuals receiving healthcare services	Own records, booking
Receptionist	Front-desk staff managing appointments and registration	Appointments, registration

2.4 Operating Environment

- Server OS: Ubuntu 20.04 LTS or Windows Server 2019+
- Runtime: Node.js v18+ with Express.js v4.x | Database: MongoDB 6.x / MySQL 8.x
- Client: Chrome 100+, Firefox 100+, Edge 100+, Safari 15+ (desktop and mobile)
- Network: HTTPS with TLS 1.2+ | Minimum 2 Mbps internet connection recommended

2.5 Design Constraints & Assumptions

- System SHALL be responsive on devices 360px – 1920px (mobile to desktop)
- Passwords hashed with bcrypt (factor ≥ 12) | JWT session tokens expire after 24 hours
- Backend API SHALL follow RESTful design | Source code maintained in Git
- All HTTP communication SHALL use HTTPS (TLS 1.2+)
- Users have reliable internet | Hospital staff trained before deployment
- Automated daily database backups | Multi-language: English and Hindi at launch

3. Functional Requirements

3.1 Patient Management Module

This module allows adding, updating, deleting, and viewing patient information.

Register new patient with Name, DOB, Gender, Contact, Address, Emergency Contact
Validate mandatory fields; reject duplicate entries (Aadhaar/phone)
Allow authorized staff to update patient details
View complete patient history: appointments, diagnoses, and bills
Soft deletion of records — archived, not permanently deleted
Search by name, patient ID, phone number, or disease

3.2 Doctor Management Module

Admin can manage doctor information including specializations, qualifications, and availability.

Add doctor with Name, Specialization, Qualification, Contact, and Availability
Update doctor information and availability schedule
Searchable doctor list filterable by specialization
Deactivate a doctor account without deleting associated records

3.3 Appointment Scheduling Module

Patients can book appointments. System sends notifications and shows staff duty shifts.

Book appointment with a selected doctor on available date and time slot
Prevent double-booking of a doctor's time slot
Modify or cancel appointment at least 1 hour before scheduled time
Send automated email/SMS reminders 24h and 1h before appointment
Calendar view for doctor availability and staff duty shifts

3.4 Medical Records Management Module

Doctors store and view patient Electronic Health Records (EHR). Patients can view only.

Create EHR entry: Diagnosis, Symptoms, Medications, Follow-up Date
View complete chronological medical history of assigned patients
Update treatment details; attach test reports (PDF/image upload)

Patients view own records but SHALL NOT edit them

All medical records encrypted at rest using AES-256

3.5 Billing and Inventory Module

Automate invoices for services, insurance claims, and payments. Monitor stock levels and issue alerts.

Auto-generate itemized invoices for services, insurance claims, and payments

Support multiple payment modes: Cash, Card, UPI, and Insurance

Record full or partial payments; track outstanding dues

Monitor inventory stock; issue alerts when stock falls below threshold

Generate financial and inventory usage reports exportable as PDF/CSV

3.6 Admin Dashboard Module

Centralized control panel with real-time metrics: total patients, today's appointments, active doctors, pending bills.

- View and manage all patient, doctor, and staff accounts
- Configure hospital-wide settings (departments, shift timings, modules)
- Access audit logs for all user actions with timestamps and IP addresses

3.7 Reporting Module

Custom reports on patient stats, bed occupancy, and revenue. Export data in PDF/CSV formats.

Report Type	Description	Format
Patient Statistics	Age, gender, disease category breakdown	PDF/CSV
Bed Occupancy	Ward utilization and bed availability report	PDF/CSV
Revenue Report	Billing collection and outstanding dues summary	PDF/CSV
Appointment Trends	Booking trends, no-show analysis, peak hours	PDF/CSV

4. External Interface Requirements

4.1 User Interfaces

Intuitive web dashboard for desktop/mobile with responsive design. Mobile layout adapts to all screen sizes.

- Responsive web dashboard: desktop (1024px+) and mobile (360px+)
- Role-specific dashboards for Admin, Doctor, and Patient with distinct navigation
- All forms include inline validation with descriptive error messages
- Multi-language support: English and Hindi in v1.0
- Interface complies with WCAG 2.1 Level AA accessibility guidelines

4.2 Hardware Interfaces

- Computer / Laptop with modern web browser | Internet connection required
- Minimum display: 1024×768px (desktop), 360×640px (mobile)
- Printer support for physical bills/reports via browser print function

4.3 Software Interfaces

Component	Technology / Version	Purpose
Backend Framework	Node.js v18+ / Express.js v4.x	REST API server and business logic
Primary Database	MongoDB 6.x	Document storage for patient/medical records
Relational DB (Optional)	MySQL 8.x	Structured billing and inventory data
Authentication	JWT + bcrypt	Session management and password hashing
Email/SMS Notifications	NodeMailer / Twilio API	Appointment reminders
File Storage	AWS S3 / Local MinIO	Medical document upload storage
Frontend	HTML5, CSS3, JavaScript ES6+	Client-side rendering

5. Non-Functional Requirements

5.1 Performance Requirements

- Support at least 100 concurrent users without performance degradation
- Average page load time SHALL not exceed 3 seconds on a 2 Mbps connection
- API response time for standard queries SHALL be under 500 milliseconds
- Database queries optimized with indexing on patient_id, doctor_id, appointment_date

5.2 Security Requirements

- Role-based access control (RBAC) enforced at API and UI levels
- Admin: full access | Doctor: EHR view | Patient: personal data only
- Passwords hashed with bcrypt (cost factor ≥ 12); JWT tokens expire after 24 hours
- Log all login attempts (success/failed) with timestamps and IP addresses
- Medical records encrypted at rest (AES-256); input sanitized (XSS/injection prevention)
- Complies with IT Act 2000, India (applicable data protection regulations)

5.3 Usability Requirements

- New users complete core tasks within 5 minutes without prior training
- Fully responsive and mobile-friendly interface (360px – 1920px)
- Multi-language: English and Hindi supported at launch
- Error messages descriptive and actionable in user's selected language

5.4 Reliability Requirements

- System achieves 99.5% uptime on a monthly basis
- Automated daily database backups with 30-day retention policy
- RTO (Recovery Time Objective): system recovery within 30 minutes of failure
- RPO (Recovery Point Objective): data loss SHALL not exceed 1 hour of transactions
- Data stored securely; prevents data loss through redundant storage

5.5 Scalability Requirements

- Architecture designed to scale horizontally up to 1,000 concurrent users
- Database schema supports sharding or replication for future growth
- SHOULD be containerized using Docker for easy deployment and portability

6. Conclusion

The Health Management System (HMS) v1.0 is a web-based platform built to modernize and simplify hospital and clinic operations. The key takeaways from this SRS are:

- The system covers all critical healthcare workflows — patient registration, EHR management, appointment scheduling, billing, inventory, and reporting — through a secure, role-based portal for Admins, Doctors, Patients, and Receptionists.
 - A modern three-tier architecture (Node.js + MongoDB + HTML5) ensures the system is scalable, maintainable, and accessible across desktop and mobile devices.
 - Strong security measures including AES-256 encryption, bcrypt password hashing, JWT authentication, and RBAC protect sensitive patient data in compliance with IT Act 2000, India.
 - Performance targets of 99.5% uptime, sub-500ms API responses, and support for up to 1,000 concurrent users make the system suitable for medium-to-large healthcare facilities.
 - Containerization via Docker, daily database backups, and multi-language support (English and Hindi) ensure reliable deployment, disaster recovery, and broad accessibility.
 - Features deferred to future versions — telemedicine video, insurance gateway integration, and native mobile apps — provide a clear roadmap for further development beyond v1.0.
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